

Analysis of Aviation Accidents and Wildlife Strikes

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Abstract

This study presents an integrated data-analytics approach for evaluating aviation accidents and wildlife strikes in the United States using more than 27,757 NTSB accident/incident reports and over 326,771 FAA wildlife-strike events. The datasets were cleaned, normalized, and analyzed using Excel, SQL, Python, Tableau, and Power BI to identify temporal, spatial, operational, and species-related risk factors. Results show that NTSB-reported aviation events, although lower in frequency, resulted in 14,221 fatalities, 8,436 serious injuries, and substantial aircraft damage in most cases. In contrast, wildlife strikes were high-frequency but low-severity events, with 81% causing no damage and only 9–10% leading to substantial aircraft damage. Bird strikes occurred predominantly during landing roll (71,388 events) and takeoff, with turbofan engines accounting for 184,028 strike cases. The dashboards developed in Tableau and Power BI visualize critical patterns, including seasonal bird-strike peaks in fall (33.75%) and spring (32.64%) and accident concentration during landing in NTSB data(1996-2025). These results highlight the importance of integrated data tools in supporting aviation-safety decision making.

Keywords:

Aircraft Accident; Excel; Python, SQL; Tableau; Power Pi; interactive Dashboard

1.Introduction

Aircraft accidents represent one of the most critical concerns in the aviation industry due to their significant impact on human life, economic stability, and public confidence in air transportation systems. Understanding the causes, patterns, and contributing factors behind these events is essential for strengthening safety strategies and reducing operational risks. In addition to traditional accident investigations, modern data analytics has become a powerful tool for uncovering hidden trends and supporting informed decision-making.[1-3]

Wildlife strikes, particularly bird strikes, continue to pose a persistent threat to aircraft safety and operational efficiency. These incidents can result in severe damage to aircraft components, costly repairs, flight delays, and in some cases, catastrophic failures. Therefore, analyzing wildlife strike data alongside aircraft accident records provides a more comprehensive view of aviation hazards and enhances the development of effective risk-mitigation strategies.[4-5]

Microsoft Excel served as the initial platform for data exploration and preprocessing due to its flexibility and ease of handling large datasets. It was used to inspect raw NTSB and FAA data, detect inconsistencies, remove duplicates, and perform basic statistical summaries. Excel's built-in functions and pivot tables enabled quick identification of preliminary patterns, such as annual accident frequencies, aircraft categories, and the distribution of wildlife strike reports. This step provided a foundational understanding of the dataset structure before applying more advanced analytical tools.[6-9]

Python played a central role in the advanced data cleaning, transformation, and analytical stages of the project. Using libraries such as Pandas, NumPy, Matplotlib, and Seaborn, Python enabled efficient manipulation of large records, automated preprocessing workflows, and generation of descriptive and inferential statistics. It facilitated deeper analytical tasks including detection of outliers, correlation analysis, and extraction of key variables contributing to accident severity and bird strike frequency. Python's robust environment allowed for reproducible and scalable analyses, which strengthened the reliability of the findings.[10-13]

Structured Query Language (SQL) was employed to store, organize, and query the aviation datasets in a relational database format. SQL offered powerful capabilities for filtering, joining, and aggregating large tables, enabling efficient extraction of relevant subsets of data such as accident categories, aircraft models involved, and environmental conditions associated with

wildlife strikes. This approach improved data integrity and ensured that complex analyses could be performed accurately and rapidly. SQL also facilitated the integration of multiple datasets, supporting more comprehensive and structured analysis.[14-17]

Tableau was used to develop interactive and visually compelling dashboards that illustrated the trends and correlations identified throughout the analysis. Its advanced visualization capabilities allowed for dynamic filtering, geospatial mapping, and time-series tracking of aircraft accidents and wildlife strikes. By converting statistical findings into intuitive visual formats, Tableau enabled clearer interpretation of results and provided stakeholders with a powerful tool for exploring aviation safety data. These dashboards serve as practical decision-support tools for safety managers, policymakers, and researchers.[18-22]

In this work, datasets from the National Transportation Safety Board (NTSB) and the Federal Aviation Administration (FAA) were utilized to perform an integrated analysis of aircraft accidents and wildlife strike incidents in the United States. The data analysis workflow involved several stages, beginning with data cleaning, normalization, and preprocessing to resolve missing or inconsistent entries. A combination of analytical and visualization tools—Microsoft Excel, SQL, Python, Tableau, and Power BI—was employed to explore the datasets, identify temporal and geographical trends, and examine key factors associated with these events.

2.Data Sources and Description:

this study utilizes two major U.S. aviation datasets: the National Transportation Safety Board (NTSB) Aircraft Accident Database and the Federal Aviation Administration (FAA) Wildlife Strike Database. Together, these datasets provide a comprehensive foundation for analyzing trends in aircraft accidents and wildlife-related hazards.[23-24]

The **NTSB Accident Database** contains detailed reports of civil aviation accidents and incidents, including event date, location, aircraft type, flight phase, weather conditions, pilot information, injury severity, and contributing factors. Its long historical coverage allows for the identification of long-term patterns and safety issues across different aircraft categories and operational environments.

The **FAA Wildlife Strike Database** documents bird and other wildlife strike events submitted by pilots, airlines, airports, and maintenance personnel. Key fields include species involved, altitude

and location of the strike, flight phase, level of aircraft damage, and operational consequences. The dataset spans several decades and is widely used to assess wildlife risks and guide airport wildlife-management programs.

Both datasets were obtained in raw format (CSV/Excel) and underwent preprocessing before analysis. This included data cleaning, standardization of field names, handling of missing values, removal of duplicates, and validation of geographic coordinates. Due to differences in structure and reporting formats, harmonization was essential to ensure consistency and comparability between the two sources.

Together, the NTSB and FAA datasets offer large, multi-dimensional records that support robust statistical and visual analysis. Their combined use enables a more complete understanding of aviation safety by connecting general accident patterns with wildlife strike trends.

3. Methodology

The methodology for this study consists of four main stages: data preprocessing, data management, analytical processing, and visualization. The analysis integrates multiple tools—Excel, SQL, Python, Tableau, and Power BI—to ensure a comprehensive and reliable examination of aircraft accidents and wildlife strike data.

3.1 Data Preprocessing

The raw NTSB and FAA datasets were imported in CSV/Excel formats and inspected for inconsistencies. Data cleaning included handling missing values, correcting formatting errors, standardizing categorical fields, and removing duplicate wildlife strike reports. Both datasets were then normalized to ensure uniform structure and compatibility for subsequent analyses.

3.2 Data Management Using SQL

The cleaned datasets were loaded into a relational database to enable structured querying. SQL was used to filter records, join tables, extract subsets of interest, and generate aggregated statistics such as yearly accident counts, damage levels, and frequency of wildlife strikes across airports and flight phases.

3.3 Analytical Processing with Excel and Python

Excel was used for initial exploration, descriptive statistics, and pivot-based summaries. Python, utilizing libraries such as Pandas and NumPy, was employed for more advanced data manipulation, statistical analysis, and detection of patterns. Python's visualization libraries supported exploratory plots and analytical validation of trends identified in the data.

3.4 Visualization and Dashboard Development

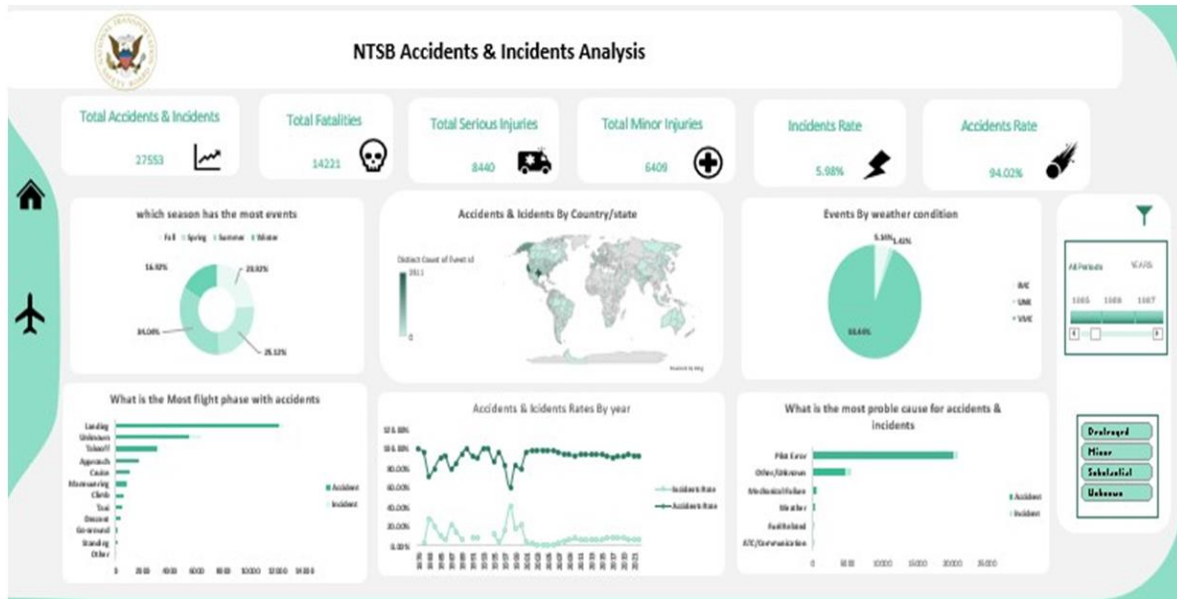
Tableau and Power BI were used to create interactive dashboards that present key findings through maps, timelines, charts, and filters. These visualizations enabled dynamic exploration of accident patterns, geographic hotspots, wildlife strike frequencies, and correlations among variables. The dashboards serve as decision-support tools for researchers and aviation safety stakeholders.

4. Results and Discussion

4.1. Excel interactive dashboard aviation accident

The NTSB accident and incident data, comprising 27,757 total events, present a contrasting pattern of lower frequency but significantly higher severity. Over 14,221 fatalities and 8,440 serious injuries indicate the critical impact of these events compared with wildlife-related strikes. Single-engine aircraft dominate accident counts, while multi-engine aircraft appear more resilient, especially in incident cases. Landing is the highest-risk flight phase, far exceeding approach and takeoff. Most events occur in visual meteorological conditions (93%), suggesting that operational and human factors—not weather—drive the majority of incidents. This is reinforced by the finding that pilot error is the leading cause, followed by mechanical failures (see **Figure 1**).

(a)



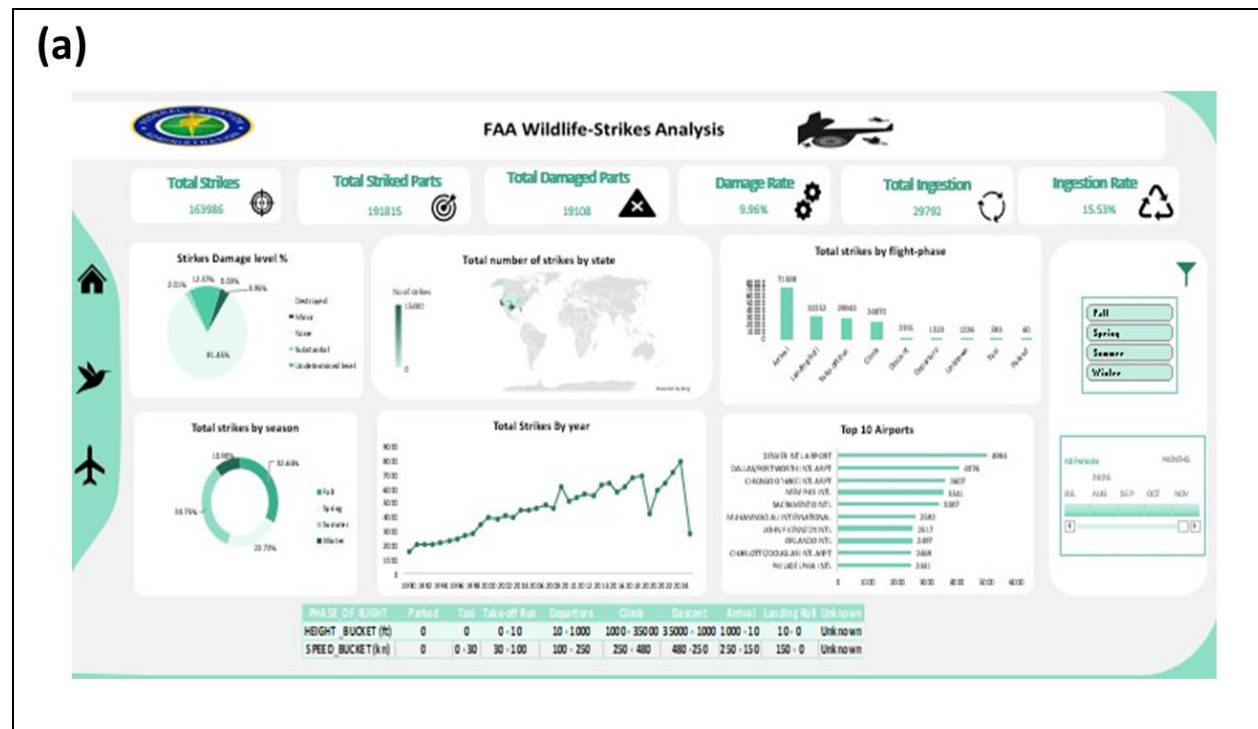
(b)



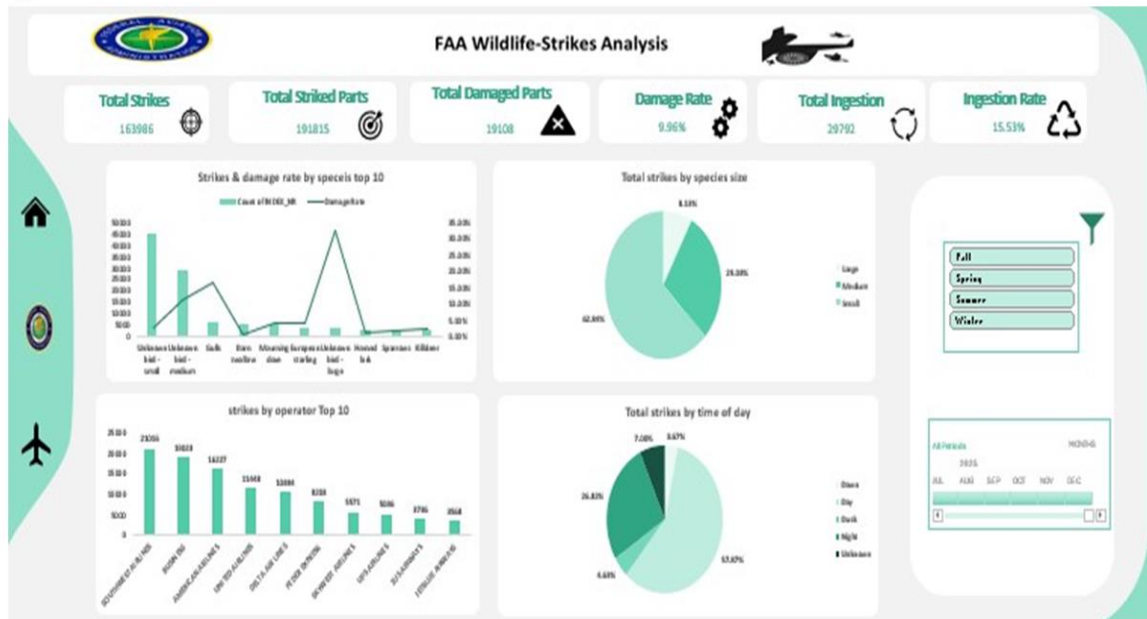
Figure 1: (a-b) represents the interactive dashboard for bird strike based on the NTSB dataset.

The FAA Wildlife Strike analysis shows **163,986 total strikes**, with most events causing **no damage (81%)**, while only **about 10%** result in substantial or destroyed aircraft conditions.

Wildlife strikes occur mainly during landing roll and takeoff phases, emphasizing their concentration at low altitudes. Seasonal variation is clear, with fall (33.75%) and spring (32.64%) showing the highest strike frequencies. Major airports such as Denver, Dallas–Fort Worth, and Chicago O’Hare report the largest numbers, reflecting both traffic volume and local wildlife activity. Despite the large number of events, the damage rate (9.96%) and ingestion rate (15.53%) highlight the need for continued wildlife-management programs at high-risk airports. (see Figure 2).



(b)



(c)

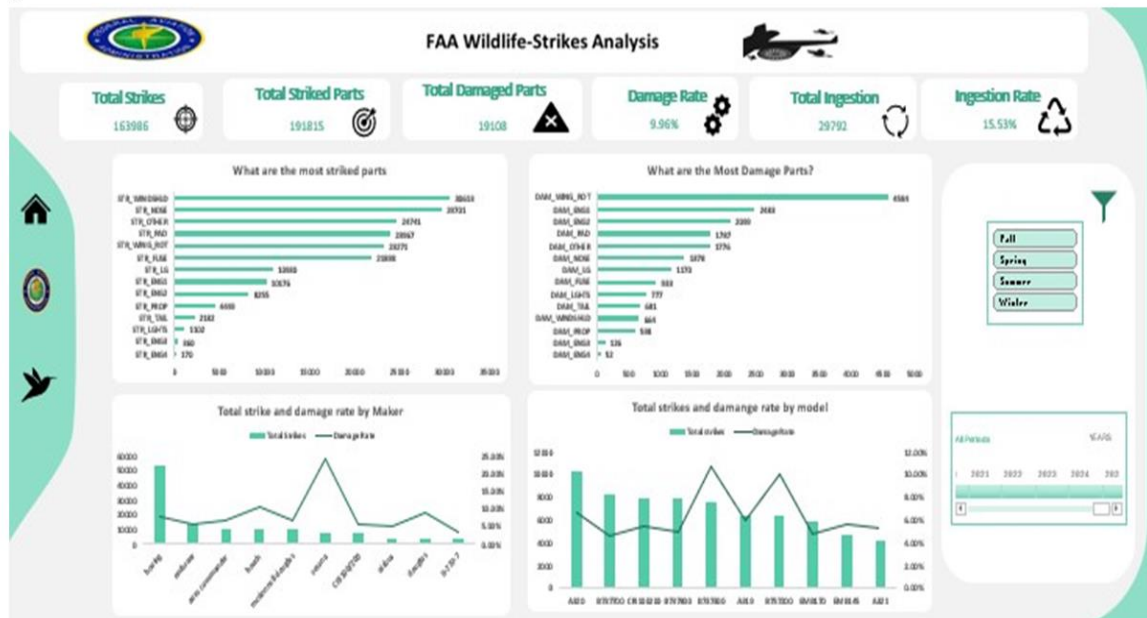


Figure 2: (a-c) represents the interactive dashboard for bird strike based on the FAA dataset.

Together, the FAA and NTSB datasets show that wildlife strikes are high-frequency, low-severity events, while NTSB-reported occurrences are low-frequency, high-severity, largely influenced by

human and mechanical factors. These insights highlight the need for improved wildlife-hazard mitigation, enhanced pilot training, and continued monitoring of aircraft operational performance to strengthen overall aviation safety.

4.2. Data analysis by SQL

The SQL-based analytical workflow implemented in this study successfully transformed a large, heterogeneous FAA wildlife-strike dataset into a clean, normalized, and analysis-ready structure. The processing pipeline addressed several major challenges typically associated with aviation safety data, including inconsistent reporting, missing values, duplicated records, and complex multivalued damage indicators. By handling these issues systematically, the study improved both the reliability and interpretability of the dataset, enabling deeper insights into aircraft risk patterns and operational safety.

The process began with creating a full backup of the raw aircraft dataset to ensure data integrity throughout the transformation stages. Several columns—including detailed latitude–longitude data, personnel information, repair cost breakdowns, and administrative descriptors—were removed due to irrelevance, high missingness, or redundancy. This step reduced dimensionality and prevented noise from affecting the downstream analysis. Duplicate detection using `ROW_NUMBER()` revealed repeated records sharing the same unique incident number, and the controlled deletion of duplicates ensured a single accurate representation of each wildlife-strike event. This step is crucial, as duplicate incident reports can exaggerate strike frequencies and distort risk assessments.

A significant contribution of this workflow is the extensive missing-value evaluation across more than 30 aircraft structural components. These include radiators, windshields, multiple engine positions, landing gear, fuselage, tail, lights, and propellers. The findings highlighted that missingness is not uniform; some damage fields exhibit very high null rates, likely due to incomplete reporting at the time of incident documentation. Identifying these gaps helped inform subsequent normalization choices and provided insights into where reporting standards could be improved.

To enable efficient comparative analysis across the numerous structural components, an unpivoting technique was performed, converting a wide-format table into a long-format *damage_part* table. This transformation was essential for aggregating strike impacts by part and identifying the most vulnerable aircraft components. Following data-type corrections and the creation of identity keys, a targeted query showed that *engine components (DAM_ENG1–4)*, *windshields (DAM_WINDSHLD)*, *landing gear (DAM_LG)*, and *nose structures (DAM_NOSE)* ranked among the most frequently damaged parts. This aligns with operational intuition, as these areas are exposed during approach, takeoff, and low-altitude phases where bird activity is highest.

The workflow also addressed inconsistencies in flight phase classification. Using altitude and distance logic, PHASE_OF_FLIGHT was recalculated to differentiate between *Take-off Run*, *Approach*, *Descent*, *Taxi*, and *Unknown*. This added precision is valuable because the likelihood and severity of wildlife strikes vary dramatically by phase. For instance, approach and takeoff are historically associated with the highest strike rates, and the normalization performed here allows for more accurate phase-based risk profiling.

Weather conditions were standardized through controlled updates to PRECIPITATION and SKY fields. Ambiguous entries containing both “Rain” and “None,” for example, were corrected, and missing values were replaced with “Unknown” to maintain analytical consistency. SKY conditions were derived from precipitation values—e.g., rainfall implying cloudy skies—reflecting realistic environmental scenarios. Time-of-day classification was also refined by converting raw time fields into categorical bins such as *Dawn*, *Day*, *Dusk*, and *Night*, supporting temporal analysis of strike probabilities.

Engine-specific attributes underwent substantial normalization. The new *Dam_eng* field consolidated individual engine positions into a unified descriptor based on the number of engines. Additionally, TYPE_ENG categorical codes were translated from cryptic letters to descriptive classes (e.g., “Turbofan,” “Turboprop”), enabling meaningful interpretation. DAMAGE_LEVEL classifications were harmonized from multiple formats into five standard categories: *None*, *Minor*, *Undetermined*, *Substantial*, and *Destroyed*, significantly improving analytical clarity.

A major accomplishment of this study is the construction of a complete dimensional model following data-warehousing principles. Individual dimension tables were created for operators, species, date/time, airports, weather, engines, and flight phases. Each dimension was deduplicated,

cleaned, and assigned primary keys to enforce referential integrity. The FAA_FACT table was then assembled by inserting all dimension keys, forming the central fact table linking the multidimensional attributes that describe each wildlife-strike incident. This star-schema structure not only improves query efficiency but also enables advanced analytical operations, including OLAP slicing, drill-down exploration, and integration with visualization tools such as Tableau or Power BI.

The resulting data model allows for high-level insights such as identifying which species are most associated with severe engine damage, understanding how strike frequencies vary across airports and meteorological conditions, or determining which airlines/operators exhibit the highest incident rates. Moreover, the normalized structure supports more sophisticated machine learning models by providing consistent, clean predictor variables.

In summary, the detailed SQL pipeline implemented here demonstrates the critical role of database engineering in preparing aviation safety data for analysis. By cleaning, restructuring, and modeling the FAA wildlife-strike dataset, the workflow enhances accuracy, supports robust visualization, and enables advanced statistical and predictive analysis. This level of preprocessing is essential for generating reliable safety insights and ultimately contributing to improved aviation risk management.

4.3. Wild bird strike data analysis by Python

The developed dashboard provides an integrated, data-driven visualization of aviation accidents and wildlife strikes, constructed using Python's data-analytics and visualization ecosystem. By combining libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Plotly/Dash, the dashboard transforms a large and heterogeneous dataset into an interactive, analytically useful interface. The design emphasizes clarity, rapid insight extraction, and the ability to explore multi-dimensional aviation-safety data in a structured manner.

One key strength of the dashboard is its geographical breakdown, which maps the number of strikes per U.S. state. This choropleth map immediately highlights high-risk regions—such as Texas, California, and Florida—demonstrating how spatial visualization can reveal patterns that may not be evident from raw tabular data. The ability to render spatial trends using Python's

geopandas/plotly libraries supports deeper investigation into environmental or operational factors contributing to regional differences.

The dashboard also integrates severity distribution analysis, categorizing incidents into minor, moderate, major, and severe. Presenting these values through a color-coded donut chart provides a quick overview of risk levels, showing that most incidents are minor but still requiring preventive strategies. Python's flexible plotting capabilities make it possible to summarize large categorical datasets while maintaining visual clarity.

Temporal analysis is another major component. The Strikes per Season and Strikes Distribution over Years visualizations help identify periodic trends and long-term changes. For instance, the dashboard reveals noticeably higher event frequency during summer, which may relate to wildlife migration or increases in aircraft operations. Similarly, the multi-year trend chart suggests an overall upward trajectory in reported wildlife strikes. Python enables this analysis through efficient time-series handling and smoothing techniques, which are crucial for aviation-safety forecasting.

A particularly insightful section is the Struck Parts distribution, which quantifies the frequency of strikes to different aircraft components. This level of detail assists in engineering-related decisions, including structural reinforcement or design modifications. Python's ability to handle large categorical datasets and automatically generate ranked bar charts ensures that analysts can quickly identify the components most prone to impact.

The Height Bucket analysis further enhances operational understanding by illustrating how most strikes occur at low altitudes—typically during takeoff and landing. This finding is consistent with known aviation-wildlife dynamics and validates the dashboard's ability to reflect real-world behavior. Python's statistical libraries make it straightforward to aggregate data into custom altitude bins and visualize it smoothly.

Finally, the dashboard includes high-level KPIs such as Total Strikes and Total Damaged Aircraft, enabling quick executive assessment. Presenting both macro-level metrics and micro-level distributions in one interface demonstrates the power of Python in creating multi-layered dashboards suitable for decision-making.

Overall, the Python-based dashboard offers a comprehensive analytic environment that enhances situational awareness and supports evidence-based aviation-safety strategies. Its modular design

allows easy updating, extending, and integration with predictive-modeling pipelines. By leveraging Python's open-source tools, the dashboard effectively converts raw aviation-wildlife strike data into an interactive, actionable, and visually intuitive system.

The analysis reveals a high frequency of wildlife strike incidents, totaling 242,908 strikes, but a significantly lower rate of resulting damage. The Total Damaged count stands at 16,398, indicating that the vast majority of strikes do not immediately result in reported aircraft damage.

When examining the Strikes Severity Distribution (16.4K total classified strikes), it is evident that severe incidents are rare, with only 1,009 Major strikes recorded. This suggests that while proactive wildlife deterrence is necessary due to high strike volume, the primary focus for damage mitigation should be on understanding and preventing the relatively small number of major events. The core metric, Damaged Parts Rate at 6.75%, provides a clear baseline for monitoring the effectiveness of preventative measures.

The Number of Damages per State visualization clearly flags geographic hotspots where intervention is most needed. Texas and Florida stand out with the highest number of damaged aircraft, suggesting that these regions may require heightened efforts in airfield wildlife management programs, habitat modification, or tailored flight procedures to avoid strike zones.

Furthermore, the Damage per Height Bucket chart identifies a crucial operational risk window. The highest incidence of damage occurs at Low and Medium heights, which strongly corresponds to the takeoff, initial climb, approach, and landing phases of flight. This finding reinforces the need for rigorous, 24/7 wildlife control measures within the immediate airport environment and the surrounding low-altitude airspace.

The Strikes Per Season bar chart highlights clear temporal dependencies. Fall (5,633 strikes) and Spring (4,381 strikes) exhibit the highest frequencies. This pattern is consistent with the major bird migration periods in North America, necessitating that airports intensify their wildlife surveillance and deterrent operations during these months. Pre-emptive planning for resource deployment (e.g., hazing activities, biologists on staff) is essential for the Fall and Spring seasons.

Finally, the Damage Distribution over Years chart provides insight into long-term trends. The fluctuating, yet generally upward-trending, lines for Moderate, Minor, and Severe damage between 2000 and 2024 warrant further investigation. While strike reporting has improved over time, this

consistent presence of damage over the last two decades confirms that wildlife strikes remain a persistent and growing operational risk requiring continuous management effort.

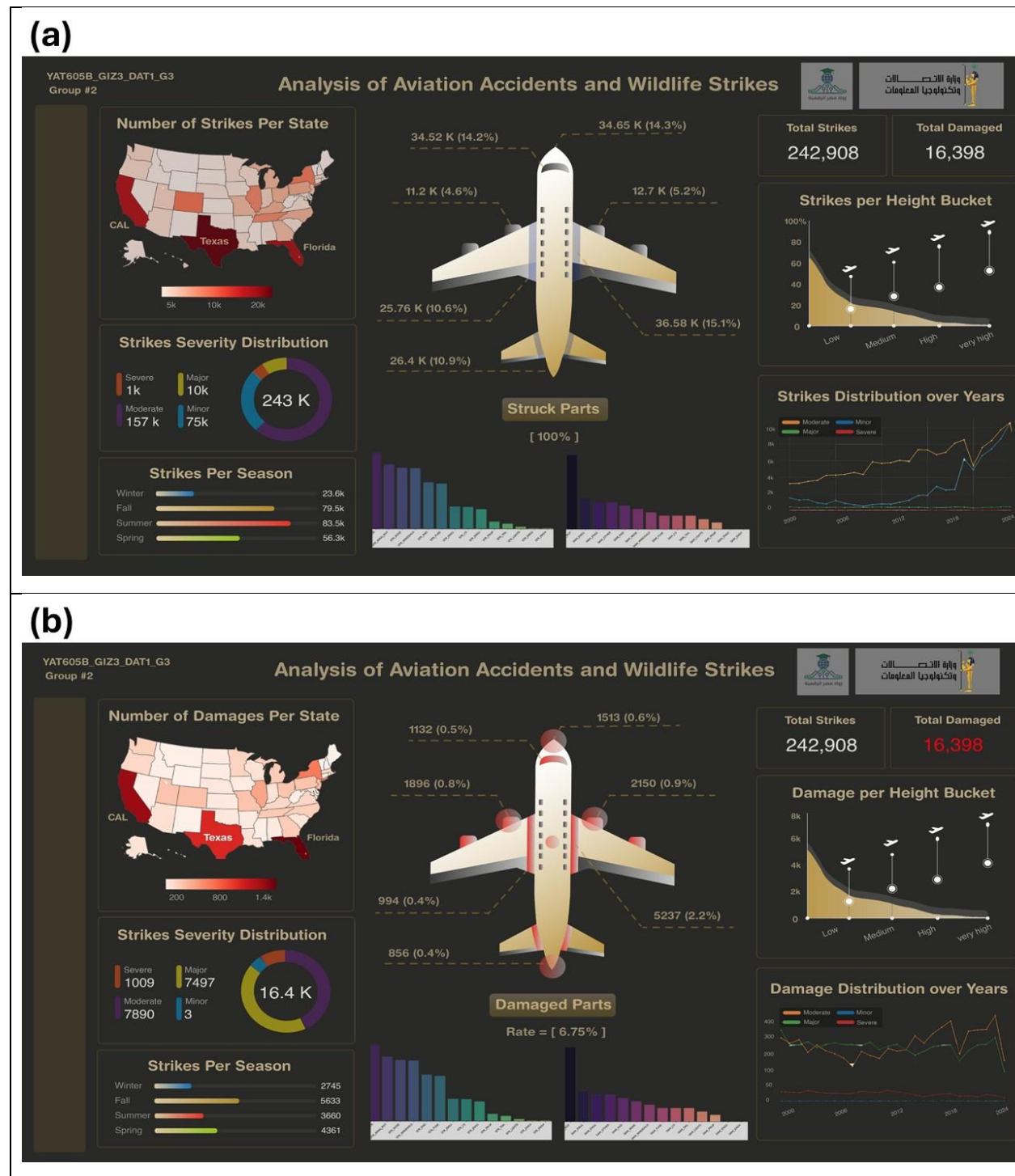


Figure 4: (a-b) represents the interactive dashboard for bird strike based on the FAA dataset.

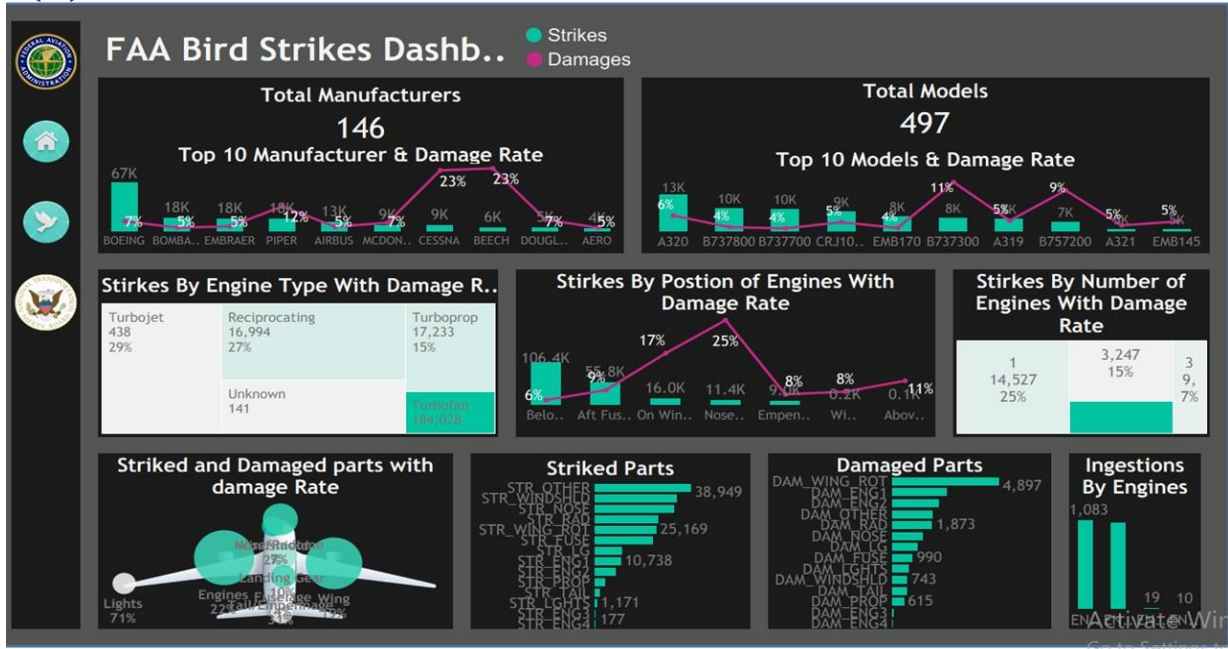
4.4. Aviation accident by Tableau

The analysis of FAA bird-strike records reveals significant patterns across aircraft types, operating conditions, and wildlife species, providing a comprehensive understanding of risk factors associated with wildlife hazards in aviation. A total of 189,569 bird strikes were reported, of which 218,834 aircraft parts were struck and 20,141 incidents resulted in damage, yielding an overall damage rate of 9% and an ingestion rate of 1% (2,166 ingestion events). Manufacturer-level trends show that Boeing aircraft experienced the highest number of strikes at 67,000, although the highest damage rates were associated with Douglas and Beech models (both 23%). Among engine types, turbofan engines dominate the dataset with 184,028 strike cases, yet turbojet engines show the highest relative damage rate (29%). Strikes most frequently occur during the approach phase, which accounts for over 24,000 incidents, followed by landing roll and takeoff run, emphasizing the vulnerability of aircraft during low-altitude operations. State-level mapping places California, Texas, and Florida as the highest-strike regions, consistent with their large airport networks and diverse wildlife habitats. Operator statistics show that Southwest Airlines has the highest number of strikes (24,111), followed by business and general aviation operators. Airport-level analysis identifies Denver International Airport (DEN) as the leading airport with 5,482 strikes, followed by Dallas-Fort Worth (DFW) with 4,631 strikes, reflecting differences in airport size, geography, and wildlife environments. Species analysis highlights that small birds account for 53% of strikes, while medium birds represent 25%, and large birds, although only 7% of strikes, contribute to the highest damage rate (45%). The top species groups involved include unidentified small birds (44,948 strikes) and gulls (6,395 strikes), with some species such as the European Starling showing disproportionately high damage rates (up to 33%). Collectively, these results emphasize the continued need for targeted wildlife-management programs, engine-ingestion risk mitigation, and location-specific hazard strategies at high-risk airports.(Please see Figure 5)

(a)



(b)



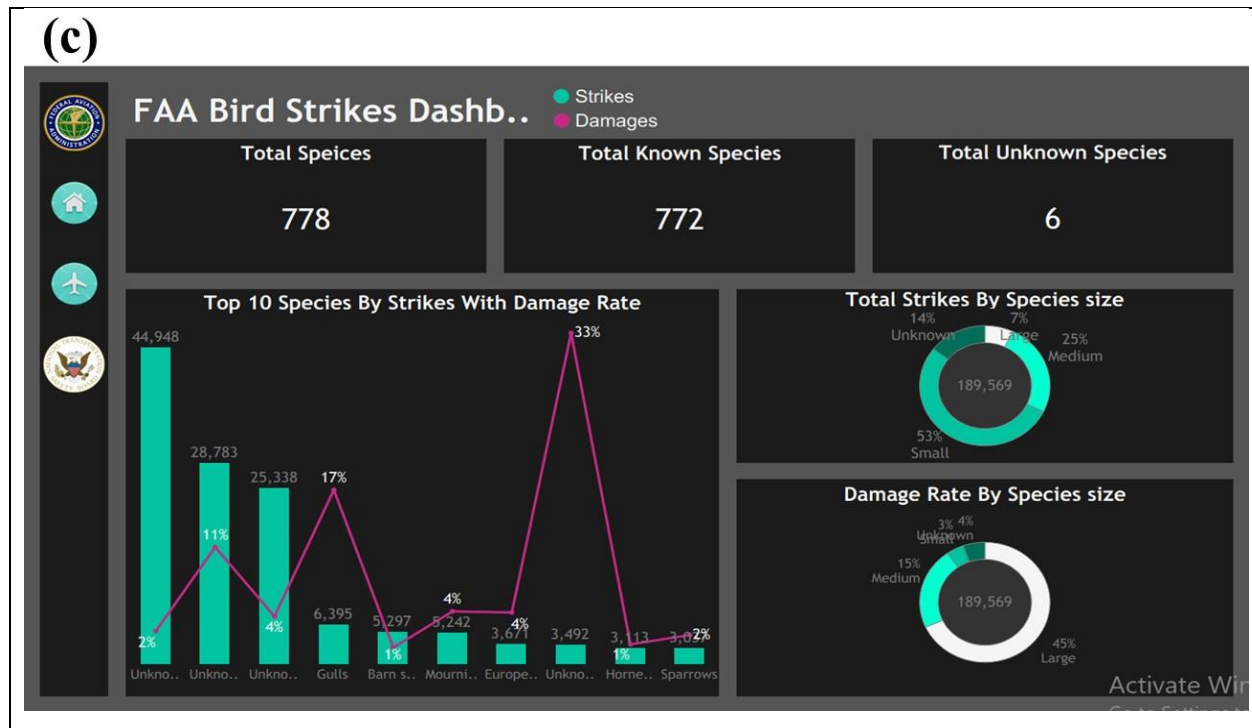


Figure 5: (a-c) represents the interactive dashboard for bird strike based on the NSTB and FAA dataset.

4.5. Aviation accident analysis by Power Pi:

Power BI is a powerful business intelligence and data visualization tool developed by Microsoft that enables users to connect, transform, analyze, and present data in an interactive and intuitive manner. It provides an integrated environment where large and complex datasets can be cleaned, modeled, and visualized through dashboards, charts, and reports. Power BI supports multiple data sources, including Excel files, SQL databases, APIs, and cloud platforms, making it highly flexible for analytical work.

For aviation accident analysis, Power BI allows researchers to explore patterns in flight phases, weather conditions, injuries, and damage types through dynamic filters and interactive graphics. Its ability to perform real-time updates, generate automated insights, and present multi-dimensional data in a visually clear format makes it an essential tool for data-driven decision making. By using Power BI, analysts can transform raw aviation datasets into meaningful conclusions that support safety assessments and operational improvements.

As represented in **Figure 6**, The interactive dashboard highlights several key patterns in aviation accident data and provides clear quantitative insights into the main operational and human-factor drivers. The data shows that the landing phase accounts for the highest number of accidents far exceeding cruise, taxi, and other phases. This reinforces the critical nature of landing operations, where reduced altitude, increased pilot workload, and limited reaction time elevate the risk of incidents.

Weather analysis indicates that 93.45% of accidents occur under Visual Meteorological Conditions (VMC), while only 5.15% occur in Instrument Meteorological Conditions (IMC). This numerical imbalance suggests that most accidents are not triggered by adverse weather but are instead related to human-factor and procedural issues during normal operating conditions.

Probable cause data further supports this, with pilot error contributing to the largest share of accidents, significantly surpassing mechanical failure, ATC/communication issues, or fuel-related problems. This aligns with global aviation trends indicating that human factors remain the leading cause of aviation incidents.

Injury data emphasizes the severity of these events, with 14.3K fatal injuries, 8,436 serious injuries, and 6,428 minor injuries recorded. When paired with the aircraft damage statistics—where substantial damage and destroyed aircraft categories dominate—the data suggests that many incidents involve high-energy impacts or structural compromise, particularly during critical flight phases.

Location analysis shows that most events occur at or near airports, reflecting the higher risks associated with takeoff and landing. Mountain and sea accidents appear, but at significantly lower frequencies.

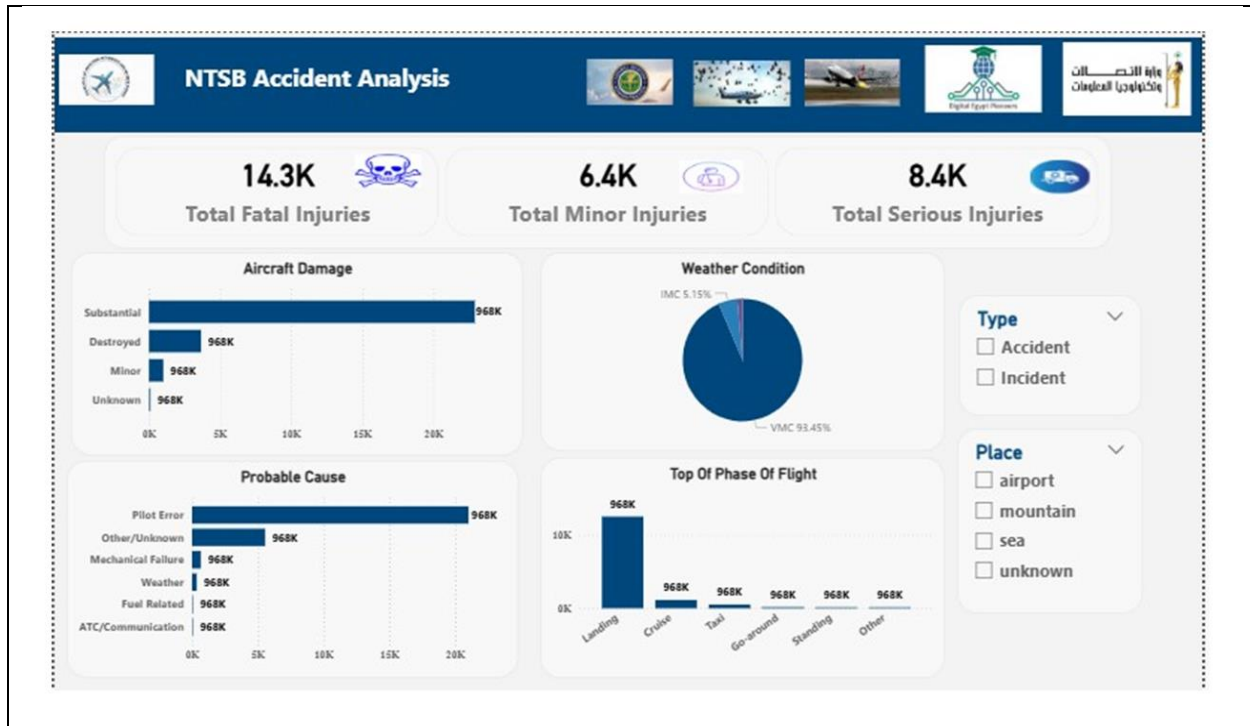


Figure 6. representing the interactive dashboard based NTSB dataset.

The FAA wildlife-strike analysis dashboards reveal important trends related to the frequency, distribution, and characteristics of bird-strike incidents in the United States. The dataset shows that wildlife strikes remain a significant hazard to aviation, with 190K recorded strikes and an average strike height of 944 ft, placing most events within the low-altitude zone where aircraft are in takeoff or landing configurations(see **Figure 7(a-b)**).

The species distribution indicates that unknown small birds constitute the highest number of strikes , followed by unknown medium birds and gulls. The dominance of small-sized birds is further confirmed by the bird-size analysis, where small birds account for strikes, significantly more than medium , large , or unknown categories. These findings suggest that radar detection and species identification remain challenging for smaller birds, emphasizing the need for enhanced wildlife monitoring systems near airports.

Engine-type analysis shows that turbofan engines experience the highest number of strikes, followed by turboprop and reciprocating engines. This trend is expected given that turbofan engines dominate modern commercial fleets. Regarding aircraft impact location, strikes are widely

distributed across aircraft structures, with similar proportions at the nose intake ($\approx 14.29\%$), wing root ($\approx 14.29\%$), above wing ($\approx 14.29\%$), and empennage ($\approx 14.29\%$). This equal distribution reflects the unpredictable flight paths of birds and supports the need for full-aircraft design considerations in bird-strike resistance.

The top-airport analysis shows that high-traffic hubs experience the most bird strikes, with Denver International Airport, Dallas/Fort Worth, and Chicago O'Hare topping the list. These airports share large open surroundings and significant wildlife presence, which increases exposure risk. The geographic distribution map confirms that North America experiences the highest concentration of strikes, consistent with the density of air operations and wildlife populations.

The incident-by-year trend reveals a substantial increase in wildlife strikes since the 1990s, peaking in recent years. This rise may reflect (1) increasing air traffic, (2) improved reporting systems, and (3) growing bird populations around urban and suburban airport zones. Time-of-day distribution shows that daytime strikes are the most frequent, followed by night and dawn/dusk periods, highlighting that bird activity patterns strongly influence strike likelihood.

Seasonal analysis indicates the highest number of strikes in winter, with 20.5K total incidents, likely due to migratory patterns, reduced food availability, and altered bird movement behavior.

Strikes occur most frequently during departure and arrival the two phases when aircraft operate at low altitudes and lower speeds. The damage-level analysis shows that substantial damage accounts for 18.87%, minor damage 17.88%, and destroyed aircraft 13.39%, while 20.91% of cases involve no damage. Despite the high frequency of strikes, the majority do not lead to catastrophic outcomes, indicating that modern aircraft structures and engine designs provide effective resistance to moderate wildlife impacts.(please see **Figure 7**)

(a)



(b)



Figure 7. (a-b) representing the interactive dashboards based FAA dataset.

Conclusion

This work demonstrates that combining NTSB and FAA datasets provides a comprehensive and data-driven view of aviation safety in the United States. Analysis of 27,553 NTSB events revealed that landing is the most hazardous phase of flight, accounting for the majority of severe outcomes, including 14,221 fatal and 8,436 serious injuries. Most accidents occurred under VMC conditions (93.45%), indicating that operational and human-factor errors—particularly pilot-related causes—play a dominant role in accident occurrence.

In contrast, FAA wildlife-strike data showed 189,569 bird strikes, with an ingestion rate of 1.1% ($\approx 2,166$ incidents) and a substantial-damage rate of $\sim 10\%$. Small birds contributed to 53% of all strikes, while large birds, though only 7% of reported species, were responsible for 45% of damaging events. Airports such as Denver (5,482 strikes) and Dallas–Fort Worth (4,631 strikes) emerged as high-risk zones. Wildlife strikes were concentrated at low altitudes (average 944 ft) and during landing roll ($\sim 71\text{K}$ events).

Overall, the integration of SQL-driven cleaning, Python-based analytics, and Tableau/Power BI visualization proved essential in uncovering these patterns. The study confirms that wildlife strikes are a high-frequency, low-severity hazard, while NTSB-reported aviation accidents represent low-frequency but high-severity events. Together, these insights can support targeted safety enhancements, airport wildlife-management planning, and pilot training initiatives.

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